

Photonics Breakthroughs 2024: Nonlinear Photonic Computing at Scale

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Abstract—A photonic neural network utilizes photons instead of electrons to process information, with the prospect of higher computing efficiency, lower power consumption, and reduced latency. This paper reviews several recent breakthroughs in large-scale photonic neural networks incorporating nonlinear operations. Specifically, we highlight our recent work, which leverages multiple light scattering and second harmonic generation in a slab of disordered lithium niobate nanocrystals for high-performance nonlinear photonic computing. The interplay of these optical effects not only enhances the computational capabilities of photonic neural networks but also increases the number of photonic computing operations. In addition, we discuss current challenges and outline future directions of nonlinear photonic computing. These advancements pave the way for exploring new frontiers in optical computing, unlocking opportunities for innovative experimental implementations, broad applications, and theoretical foundations of photonic neural networks.

Index Terms—Machine learning, neural networks, nonlinear optics, optical computing.

I. INTRODUCTION

THE growing demand for computing power drives the pursuit of next-generation neuromorphic and other computing hardware. Among various emerging platforms, photonics has become a viable solution thanks to some of the exceptional properties of light including high parallelism and low-loss transmission [1]. The field of photonic computing, which dates back to 1980s, has been revived in recent years due to the increasing need for energy-efficient, high-throughput, and cost-effective computation [2]. Additionally, advancements in photonic technologies – driven by progress in optical telecommunications and imaging – have provided a versatile toolbox for photonic computing. Meantime, the prevalence of deep neural networks in science and technology highlights the urgent need for more efficient implementations of these models. This has enabled diverse implantations of photonic neural networks (PNNs), encompassing systems based on free-space optics, fiber optics to integrated photonics, for a wide range of machine learning applications [2].

To obtain superior computing performance, deep learning has become the standard approach in modern machine learning [3]. Deep neural networks generally comprise multiple layers, with nonlinear activation functions applied between them. Similarly, nonlinearity is also a prerequisite for PNNs to implement multiple layers, thus enhancing their computational capabilities. In optics, several methods have been developed to implement nonlinear activation functions, such as square-law detector nonlinearity [4], electro-optic and opto-electronic nonlinearities [5], [6], and electromagnetically-induced-transparency nonlinearity [7]. However, these nonlinearities generally convert optical waves into other formats like electrical signals, posing challenges for subsequent photonic processing. Optical nonlinearity is thus highly desirable for all-optical or quasi-all-optical processing [8], [9], [10]. A paradigm demonstration leverages the third-order nonlinearity of silica multimode fibers for nonlinear photonic computing [11]. Yet, the scale of nonlinear computation is constrained by the number of modes supported by the waveguides, and is fundamentally limited by the tradeoff between the computing scale and energy efficiency.

To this end, more efficient nonlinear photonic computing may be realized with a lower-order nonlinearity, i.e., the second-order

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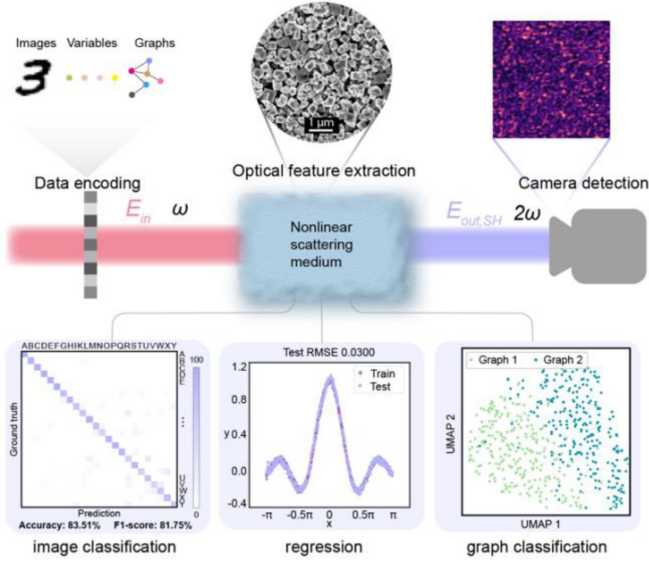


Fig. 1. Large-scale photonic computing with a nonlinear scattering medium (a slab of disordered lithium niobate nanocrystals). The dataset (e.g., images, variables, and graphs) are encoded into the phase front of the input light beam via a spatial light modulator. After propagating through the medium, second-harmonic speckle pattern is generated and measured by a camera. The second-harmonic speckles are used as features for image classification (classifying American sign language alphabet), regression (interpolating a nonlinear sinc function), and graph classification (classifying two-category synthetic graphs). Figure is adapted from ref. [12].

optical nonlinearity. In our work, we have demonstrated a large-scale PNN that combines second-order nonlinearity and multiple light scattering for various machine learning applications [12]. The computing core is a slab of disordered lithium niobate (LN) nanocrystals [13]. Each nanocrystal, when excited by a pump beam at the fundamental harmonic (FH), generates second-harmonic (SH) waves with random amplitude and phase. These SH waves interfere within the LN slab, giving rise to global SH emission, a phenomenon termed random quasi-phase-matching [14]. As a consequence of the complex interference, the output light forms a seemingly random pattern, known as *speckle pattern*. By encoding the input information into the wavefront of the pump beam, the SH generation nonlinearly transforms the input into SH speckle patterns, which serve as optical features that are used to address machine learning tasks. Compared to the linear speckle features that are also naturally computed in our system through the linear scattering of the pump beam, the nonlinear optical features exhibit improved performance across diverse datasets of varying complexity. With up to 27648 input and 3500 nonlinear output modes demonstrated in the experiment, our PNN achieves large-scale nonlinear computations all-optically.

II. BREAKTHROUGH RESEARCH

Fig. 1 illustrates the schematic setup of our nonlinear PNN. It mainly consists of four parts: data encoding, optical feature extraction, intensity detection and digital readout. Here, optical features are extracted via the SH generation process in a disordered LN slab, which is assembled from LN nanocrystals via drop deposition and solvent evaporation [13]. The input dataset is encoded into the femtosecond light pulses via a spatial light

modulator (SLM), while the generated SH speckle is measured with a camera. A simple ridge regression is applied to the SH speckle features for digital readout. The whole system can be used to compute optical features of versatile machine learning tasks in image classification, regression, and graph classification.

Fig. 1 highlights three representative datasets we studied with our PNN, i.e., classifying American sign language alphabet (24-category), interpolating a nonlinear sinc function, and classifying a synthetic graph dataset generated by the stochastic block model (2-category). High classification and regression accuracies are achieved in the experiment. More importantly, results derived from SH speckle features consistently outperform their linear counterpart in nearly all the datasets examined. These findings unequivocally underscore the critical role of optical nonlinearity in complex PNNs.

The scale of our PNN can be inherently large. The input and output dimensions are given by the mega-pixel SLM and camera, respectively, both of which are commercially available. For linear scattering media, light propagation through the media corresponds to matrix multiplication with a random matrix of dimension $10^6 \times 10^6$ [15]. The linear relationship between input field E_{in} and output field $E_{out, FH}$ is modelled as $E_{out, FH} = W_l E_{in}$, where W_l is called transmission matrix. When it comes to nonlinear scattering media, such as in our case with a LN disordered medium possessing second-order nonlinearity, the scale of computation is even larger, which can be represented by a tensor product of size $10^6 \times 10^6 \times 10^6$ [16] or an exotic two-layer neural network [12]. With tensor product representation, the nonlinear neural network is formulated as $E_{out, SH} = W_{nl} \text{vec}(E_{in} \otimes E_{in})$, where $\otimes$ and ‘vec’ denote the outer product operation and the vectorization of a matrix, respectively. W_{nl} is a matrix reshaped from a tensor that characterizes the nonlinear scattering process [16]. Such rich computation underpins the excellent performance of our experimental results, enabled by strong optical nonlinearity and the large dimensionality of our system [17]. The improved performance of nonlinear scattering over linear scattering is consistently observed across a wide range of machine learning tasks [12].

In addition to our work, there have been many other investigations in implementing and improving nonlinear PNNs. For instance, tailoring third-order nonlinear interactions of light fields in multimode fibers have been demonstrated to enhance the computational expressivity [18]. For PNNs utilizing incoherent light, quantum dot films and image intensifiers have been employed as nonlinear activations for applications in image processing and sensing, respectively [19], [20]. Another innovative and increasingly popular strategy to designing nonlinear PNNs is the concept of structural nonlinearity [21], where data itself is encoded into the system parameters rather than following the conventional paradigms of encoding it solely into the input waves [22], [23], [24], [25].

III. DISCUSSIONS AND PERSPECTIVES

Recent efforts in large-scale PNNs not only address some critical concerns but also unlock exciting opportunities in optical computing. Fundamentally, it shows that optical nonlinearity,

though differing from typical activation functions used in modern deep neural networks like ReLU or sigmoid, is beneficial for machine learning performance. It also demonstrates that PNNs can be scalable in network depth, both in theory and experiments, making them competitive in processing various tasks while accelerating neural computations. However, new challenges arise that could be pivotal for engineering the next generation of advanced PNN systems. For instance, an important direction for improving our system is to program or tune the optical computations at will. With the current technologies for data encoding, such as mega-pixel SLMs or digital micromirror devices, there is ample space to introduce additional programmable neurons to optimize nonlinear PNNs without sacrificing dimensionality [18], [26], [27]. Compared to directly fabricating a nonlinear scattering medium on demand, this method of programming PNNs is more feasible in the near term. In addition, the second harmonic generation process occurs primarily once in our system (cascaded nonlinear effects are much weaker), which limits the depth of the PNNs. To enhance the computing complexity and scalability, we could cascade the current optical computations multiple times to investigate the scaling properties.

Despite recent exciting breakthroughs, nonlinear PNNs are still in their early stages. For instance, unlike deep learning, where innovations are benchmarked against standardized datasets and metrics, there is no consensus on fair evaluation methods for PNN systems. Some pioneering attempts exist [28], [29], but fairly comparing PNNs is still elusive when jointly considering both hardware-related metrics and task-related metrics. Establishing such benchmarks is essential to standardize the development of practical optical computers. Secondly, the nonlinearity could be a double-edged sword as it often consumes more energy to excite sufficient and measurable nonlinear signals with conventional nonlinear optical mechanisms. For example, in our current platform and related ones [8], [12], the typically weak nonlinear susceptibility tensors of materials in nonlinear optics limit the energy efficiency. In other approaches that explore more efficient structural nonlinearity, the data read-in costs to create enough such nonlinearity for each inference, as opposed to the purely passive light-matter interactions, might introduce additional energy consumption. From a theoretical perspective, quantitatively understanding the origin of performance improvement brought by nonlinearity is another interesting question [30]. Such insights could lead to the advancements of more reliable optical computing hardware and open up new applications that require more explicit data features. Lastly, to fully unlock the capabilities of nonlinear PNNs in machine learning applications, synergistic design between hardware and software is indispensable. For example, techniques such as knowledge distillation and sparsity regularization have been applied to push the performance limits of linear diffractive neural networks with considerable improvements [31]. This clearly indicates that algorithmic advancements are likewise significant as hardware developments. Conversely, optical nonlinear Schrödinger equation has been reformulated as a trainable model for physics-informed neural networks, which also spark the creation of several new model architectures [32]. Similar to how diffusion models draw inspirations from non-equilibrium

statistical physics [33], this dynamical interplay between optics and machine learning might shape the long-term pathway of photonic computing.

IV. CONCLUSION

In summary, we have demonstrated a large-scale nonlinear PNN leveraging the combination of second-order nonlinearity and multiple light scattering. This approach not only enhances performance across various tasks but also significantly increases computing density. Simulating similar complex and nonlinear physical phenomena with general-purpose digital computers would require higher computational cost due to the nonlinearity and dimensionality of the PNN. As an extension, we briefly review some other related efforts in the developments of nonlinear PNNs. This rapidly evolving field holds immense potential, offering opportunities to develop more efficient and programmable optical systems, establish stronger theoretical foundations and evaluation standards, and foster deeper synergies between hardware and software.

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